

(Submission for 11th Emerging Markets Finance Conference, 2020)

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**IMPACT OF EXECUTIVE COMPENSATION ON CORPORATE TAX
AGGRESSIVENESS: EVIDENCE FROM INDIA**

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Abstract

There is growing evidence that tax aggressiveness of companies is associated with more significant incentives to their executives. The present study explores the relationship between executive compensation and corporate tax aggressiveness in India. The sample comprised a subset of the S&P BSE 500 Index companies for FY 2014-15 through 2018-19. A fixed-effects panel model was invoked with and without the moderating effect of a proxy for corporate governance strength on the executive compensation – corporate tax aggressiveness link. The results highlight that the ratio of the average variable to fixed component of the compensation in case of Indian CEOs and CFOs was 34:66 and 8:92, respectively. The econometric analysis evinced a significant negative relationship between the fixed component of executive compensation and tax aggressiveness, specifically with the moderation of corporate governance strength. In addition, no significant relationship was found in the case of variable compensation. The study provides insights on an insufficient incentive-alignment between shareholders and managers in India owing to an insignificant amount of variable pay in the total executive compensation. It puts forth vital implications for policy makers and practitioners highlighting the pertinence of the right mix of variable and fixed pay, with particular emphasis on the former, in aligning the incentives of shareholders and managers in India.

JEL classification: G34, H25, M52.

Keywords: CEO Compensation, CFO Compensation, Corporate Tax Aggressiveness, Corporate Governance.

1 Introduction

Individuals as well as corporations are required to pay a certain part of their income, as tax, to the government. Taxes, thus, constitute an important source of the governmental revenue. However, both individuals as well as corporations tend to carry out a number of activities that focus on the goal of minimising their tax liability to the government (Desai and Dharmapala, 2006; Slemrod and Yitzhaki, 2002). Such tax-aggressive activities have become a pervasive feature of many countries around the globe, particularly, in the corporate arena (Lanis and Richardson, 2011).

The term ‘corporate tax aggressiveness’ has been used in the literature, generally, for the activities reducing explicit taxes of a firm (Chen et al., 2010; Huang et al., 2018; Lanis and Richardson, 2011; Rego and Wilson, 2008; Rego and Wilson, 2012). Specifically, corporate tax aggressiveness includes typical tax planning strategies, that are perfectly legal, as well as tax avoidance activities, that are fraudulent in nature (Rego and Wilson, 2008). According to Chen et al. (2010, p. 42), tax aggressiveness “*encompasses tax planning activities that are legal or that may fall into the grey area, as well as the activities that are illegal*”.

Corporate tax aggressiveness has a positive effect on shareholders’ wealth since it leads to the generation of tax savings that accrue to the shareholders (Desai and Dharmapala, 2006). On the other hand, tax aggressive positions taken by a firm may also result in penalties imposed by the taxing authority if the firm gets audited (Campbell et al., 2017). Thus, corporate tax aggressiveness is seen analogous to risky net present value projects which are favoured by owners or shareholders who are risk-neutral but not by risk-averse managers (Rego and Wilson, 2012).

This risk-related agency problem posed by corporate tax aggressiveness requires that the owners need to put in place adequate incentives to motivate the risk-averse managers to

make tax-efficient decisions that increase their after-tax wealth (Crocker and Slemrod, 2005). Several studies, mostly in the US, have examined whether such incentive compensation contracts, primarily equity-based, encourage optimal tax outcomes and have found mixed evidence (e.g., Armstrong et al., 2012; Armstrong et al., 2015; Campbell et al., 2017; Desai and Dharmapala, 2006; Minnick and Noga, 2010; Phillips, 2003; Rego and Wilson, 2012). There is, however, a dearth of studies in the case of emerging economies where managerial compensation contracts are typically unlike that for the western parts of the world. In India, the variable component of an executive's compensation constitutes a relatively insignificant part (Chakrabarty et al., 2012; Jaiswall and Firth, 2009; Saravanan et al., 2016) with a meagre grant of stock options and that too in a few companies (Balasubramanian et al., 2010). Further, little is known about how tax aggressiveness is associated with the incentive compensation of executives.

By drawing on the agency theory, the present study aims to fill this research gap by discerning the relationship between executive compensation and corporate tax aggressiveness in the Indian context, which has not yet been explored. The compensation contracts of both the CEOs as well as the CFOs were separately taken into account in order to explore the said relationship. Further, the impact of both the fixed and variable part of the executive compensation has been studied. Specifically, the value of stock option grants as a variable component has also been analysed, which is seldom considered in the compensation literature in India. Finally, the moderating effect of the strength of corporate governance on the executive compensation – tax aggressiveness link has also been examined.

2 Review of Literature and Hypotheses Development

Corporate tax aggressiveness exhibits a risk-related agency problem (Rego and Wilson, 2012), which can lead to an increase in shareholders' wealth provided managers are risk-takers (Campbell et al., 2017). Through the projected tax savings, these activities tend to

increase the expected cash flows of a firm, however, with an increased level of dispersion of such cash flows (Goh et al., 2016). Such tax aggressive strategies are analogous to risky investment projects that are value-enhancing and are favoured by risk-neutral shareholders but not by risk-averse managers (Armstrong et al., 2015).

Agency theory predicts that the shareholders must incentivise the risk-averse managers to engage in an optimal level of tax aggressiveness (Crocker and Slemrod, 2005). This can be done explicitly based on the tax outcomes of the firm or implicitly by way of compensation contracts that link the executive pay to after-tax returns or stock price (Hanlon and Heitzman, 2010). The variable pay that may take the form of equity-based compensation or bonus and commission incentivises the managers to work in the shareholders' interests (Parthasarathy et al., 2006).

Prior research on the relationship between equity-based compensation and tax aggressiveness documents a positive relationship between the two. Studies argue that when a firm engages in risky tax strategies, the volatility of stock returns increases, which in turn increases the value of stock options granted to the executives (e.g., Armstrong et al., 2012; Armstrong et al., 2015; Minnick and Noga, 2010; Rego and Wilson, 2012). For bonus and commission, Phillips (2003) found no association, whereas Gaertner (2014) and Schmittdiel (2014) reported a positive relationship. They argued that the managers need to be compensated for taking up the excessive risk inherent in a tax strategy that is aggressive, resulting in framing of the following hypothesis:

H1a: There is a significant positive impact of the variable component of executive compensation on corporate tax aggressiveness.

The tax aggressiveness literature focuses primarily to the western parts of the world, especially the US. Huang et al. (2018) called for more studies in other international settings where managerial compensation contracts are dissimilar from that of the western part of the

globe. In India, particularly, the variable component does not constitute a significant part of the total executive pay. Chakrabarty et al. (2012) documented a massive skew in the distribution of the percentage of the variable to the total compensation given to executives in India. Also, Jaiswall and Firth (2009) found that the proportion of the median variable component to the salary of CEOs was just 10 per cent in case of the BSE 200 Index companies.

Within the variable component, stock options are not very popular amongst Indian firms (Saravanan et al., 2016). Balasubramanian et al. (2010) reported that a meagre 16 per cent of the sample S&P BSE 500 Index companies granted such options. Further, since the fixed component is mostly cash-based salary, Indian executives may have a higher level of risk aversion consistent with the theoretical argument put forward by Rego and Wilson (2012). In line with this argument, Huang et al. (2018) documented a negative relationship between executive cash compensation and corporate tax aggressiveness. Based on these studies, the following hypothesis will be tested:

H1b: There is a significant negative impact of the fixed component of executive compensation on corporate tax aggressiveness.

Modern corporations are characterised by the separation of ownership and control. The interests of the professional managers, entrusted with the decision making process in case of widely held corporations, are not identical to the interests of the owners or the shareholders of such corporations (Fama and Jensen, 1983). The implication of the divorce between ownership and control is that in case tax aggressiveness is a value-enhancing activity, then the owners need to encourage the managers to make optimal tax decisions that enhance shareholders' wealth (Crocker and Slemrod, 2005). Building on this aspect of the agency theory, specific governance mechanisms can help to align the conflicting interests of shareholders and managers (Armstrong et al., 2015). The role of institutional ownership has

long been recognised to promote good governance (Moore, 2012). In India, the institutional investors have considerable corporate shareholdings (Chakrabarty et al., 2012). Such shareholders actively monitor the management and help reduce managerial myopia (Sarkar and Sarkar, 2020; Shleifer and Vishny, 1986). Khan et al. (2017) reported a positive relationship between the level of institutional ownership and corporate tax aggressiveness.

Another implication of the ownership-control separation is that managers, in widely held corporations, may seek private gains by way of expropriating shareholders' wealth in case a divergence of interests persists (Fama and Jensen, 1983; Jensen and Meckling, 1976). Based on this agency theory perspective, Desai and Dharmapala (2006) proposed that tax aggressiveness and wealth expropriation by the self-interested managers may go hand in hand. They further explained that the obscure tax aggressive activities tend to reduce the level of corporate transparency, thus increasing the opportunity for the self-interested managers to engage in wealth expropriation for private gains. Also, according to them, well-governed firms generally have efficient internal control mechanisms in place in order to prevent such expropriation of shareholders' wealth. Therefore, the strength of corporate governance of a firm should positively moderate the relationship between executive compensation and corporate tax aggressiveness. In other words, the said relationship will be more positive or less negative in case of well-governed firms (Desai and Dharmapala, 2006). Hence, the following hypotheses have been framed:

H2a: There is a significant positive impact of corporate governance strength on the relationship between the variable component of executive compensation and corporate tax aggressiveness.

H2b: There is a significant positive impact of corporate governance strength on the relationship between the fixed component of executive compensation and corporate tax aggressiveness.

3 Data and Methodology

3.1 Sample Selection and Data Sources

The sample for the present study comprised a subset of companies listed on the S&P BSE 500 Index as on March 31, 2019. Out of the 500 companies forming a part of the index, first of all, public-sector companies were removed since their policies are influenced by a number of social obligations. Second, all banking and other financial companies were also excluded from the sample because they are governed by the Banking Regulation Act, 1949 and the RBI Act, 1934 and hence are different from the companies governed solely by the Companies Act. Third, all the companies that were not listed in the index for the entire study period were kept out of the sample. Fourth, for consistency in the data, the companies having financial year ending other than on March 31st were also dropped from the sample since they make comparisons difficult leaving out a set of 281 companies or 1,405 firm-year observations for the sample period, i.e., FY 2014-15 to FY 2018-19.

Further, out of these 1,405 observations, those characterised by a change in the CEO/CFO were omitted for uniformity (Jaiswall and Bhattacharya, 2016). Finally, observations with negative pre-tax income as well as observations with missing data to compute the variables required for the study were removed (Drake et al., 2017; Rego and Wilson, 2012; Wahab and Holland, 2012). Table 1 reports the various criteria for sample selection used in the study. For the analysis with respect to CEO (CFO) compensation, these criteria resulted in an unbalanced dataset of 255 (258) companies across 41 (42) industries, or 1,063 (1029) observations for the study period. Table 2 presents the industry composition of the sample. As shown in the table, pharmaceutical and chemical companies have the largest and the second largest industry percentage in the sample respectively. The data sources included the corporate database (Prowess) maintained by the Centre for Monitoring the Indian Economy as well as annual reports of the sample companies.

Table 1: Sample Selection

Criterion	Companies	Observations	
		(1)	(2)
S&P BSE 500 Index companies as on 31 March 2019	500		
(-) Public sector companies	(59)		
(-) Banking and financial companies	(78)		
(-) Companies not listed for the complete sample period	(49)		
(-) Companies with the financial year ending other than on 31 st March	(33)		
Preliminary Sample	281	1405	1405
(-) Observations characterised by a change in the CEO/CFO		(146)	(168)
(-) Observations with negative pre-tax income		(115)	(115)
(-) Observations with missing data to compute the variables used		(81)	(93)
Final Sample	255/258	1063	1029

Source: Authors' calculation.

Note: (1) Columns (1) and (2) denote the sample for CEO compensation and CFO compensation respectively. (2) The criteria for sample selection resulted in an unbalanced dataset of 1063 (1029) observations in case of CEO (CFO) compensation across 255 (258) companies for the study period.

3.2 Variable Selection and Description

3.2.1 Dependent variable

‘Effective tax rates’ and ‘Book-tax differences’ have been widely used in the prior literature as proxies for corporate tax aggressiveness (*TAXAGG*) (e.g., Armstrong et al., 2012; Campbell et al., 2017; Desai and Dharmapala, 2006; Dyreng et al., 2010; Gartner, 2014; Huang et al., 2018; Minnick and Noga, 2010; Phillips, 2003; Rego and Wilson, 2008; Xian et al., 2014). Effective tax rate (*ETR*) is calculated as the ratio of a tax liability estimate to the profit before tax (Hanlon and Heitzman, 2010). Following previous studies, multiple effective tax rate measures were used (Armstrong et al., 2012; Campbell et al., 2017; Chen et al., 2010; Donhoe, 2015; Dyreng et al., 2010; Gaertner, 2011; Minnick and Noga, 2010). First, the GAAP effective tax rate (*GAAP_ETR*) was calculated as the ratio of total tax expense to the pre-tax income. Second, the current effective tax rate (*CURR_ETR*) was taken as the ratio of current tax expense and the pre-tax income. Finally, the cash effective tax rate (*CASH_ETR*) was determined as the ratio of taxes paid in cash to the pre-tax income.

All the *ETR* measures were truncated to lie between 0 and 1 (Armstrong et al., 2012; Chen et al., 2010; Gaertner, 2014). Hence, observations with negative *ETRs* as well as those where the value of *ETR* exceeded 1 were treated as missing. Further, since a reduction in the

Table 2: Industry Composition of the Sample

NIC	Industry Description	(1)		(2)	
		No.	%	No.	%
1	Crop and animal production, hunting and related service activities	5	1.96	5	1.94
10	Manufacture of food products	7	2.75	8	3.10
11	Manufacture of beverages	2	0.78	2	0.78
12	Manufacture of tobacco products	2	0.78	2	0.78
13	Manufacture of textiles	6	2.35	6	2.33
14	Manufacture of wearing apparel	1	0.39	1	0.39
15	Manufacture of leather and related products	1	0.39	1	0.39
16	Manufacture of wood and of products of wood	2	0.78	2	0.78
17	Manufacture of paper and paper products	1	0.39	1	0.39
19	Manufacture of coke and refined petroleum products	1	0.39	1	0.39
20	Manufacture of chemicals and chemical products	33	12.94	33	12.79
21	Manufacture of pharmaceuticals, medicinal chemical and botanical products	23	9.02	23	8.91
22	Manufacture of rubber and plastics products	14	5.49	14	5.43
23	Manufacture of other non-metallic mineral products	17	6.67	16	6.20
24	Manufacture of basic metals	12	4.71	12	4.65
25	Manufacture of fabricated metal products, except machinery and equipment	1	0.39	1	0.39
27	Manufacture of electrical equipment	12	4.71	12	4.65
28	Manufacture of machinery and equipment	11	4.31	11	4.26
29	Manufacture of motor vehicles, trailers and semi-trailers	12	4.71	12	4.65
30	Manufacture of other transport equipment	3	1.18	3	1.16
32	Other manufacturing	3	1.18	3	1.16
34	Diversified	7	2.75	6	2.33
35	Electricity, gas, steam and air conditioning supply	2	0.78	3	1.16
41	Construction of buildings	8	3.14	7	2.71
42	Civil engineering	14	5.49	15	5.81
46	Wholesale trade, except of motor vehicles and motorcycles	8	3.14	9	3.49
47	Retail trade, except of motor vehicles and motorcycles	5	1.96	5	1.94
49	Land transport and transport via pipelines	0	0.00	1	0.39
50	Water transport	1	0.39	1	0.39
52	Warehousing and support activities for transportation	4	1.57	4	1.55
55	Accommodation	3	1.18	3	1.16
58	Publishing activities	3	1.18	3	1.16
59	Motion picture, video and television programme production	2	0.78	2	0.78
60	Broadcasting and programming activities	3	1.18	3	1.16
61	Telecommunications	5	1.96	5	1.94
62	Computer programming, consultancy and related activities	13	5.10	13	5.04
63	Information service activities	2	0.78	2	0.78
70	Activities of head offices; management consultancy activities	2	0.78	2	0.78
77	Rental and leasing activities	1	0.39	1	0.39
82	Office administrative, office support and other business support activities	1	0.39	1	0.39
86	Human health activities	1	0.39	2	0.78
93	Sports activities and amusement and recreation activities	1	0.39	1	0.39

Source: Authors' compilation.

Notes: (1) Columns (1) and (2) denote the sample for CEO compensation and CFO compensation respectively as shown in Table 1. (2) Industry classification is based on two-digit National Industrial Classification (NIC) codes given by the Central Statistics Office, Ministry of Statistics and Programme Implementation, Government of India.

ETRs denotes an increase in tax aggressiveness, all the *ETR* measures were multiplied by -1 (Drake et al., 2017; Khan et al., 2016). This was done for the ease of interpreting the results such that an increase in any of the *ETR* measures implies an increase in corporate tax aggressiveness.

Book-tax difference (*BTD*) or book-tax gap refers to the difference between the book income and the taxable income of a firm (Hanlon and Heitzman, 2010). Since it is difficult to know the actual taxable income of a firm owing to the confidentiality of the income tax returns, an estimate of the same was arrived at using the methodology proposed by Manzon and Plesko (2002). The current tax expense of a firm was grossed-up, using the highest statutory tax rate, to get an estimate of the taxable income of that firm (Armstrong et al., 2012; Moore, 2012). This measure of estimated taxable income was then subtracted from the pre-tax income in order to compute the book-tax differences (Chen et al., 2010; Desai and Dharmapala, 2006; 2009).

Desai and Dharmapala (2006) argued that the entire book-tax differences may not be attributable to corporate tax aggressiveness. According to them, earnings management may also contribute to the book-tax differences and the same needs to be controlled for while computing a proxy for corporate tax aggressiveness. Hence, following Desai and Dharmapala (2006), the *BTDs* were regressed on total accruals, a proxy for earnings management. Total accruals were computed using the cash flow approach, as the difference between the accounting income and the operating cash flows of a firm (Chen et al., 2010; Rego and Wilson, 2012). Residual book-tax difference (*RBTD*), estimated as the residual from running the regression of *BTD* on the total accruals of a firm, was taken as another proxy of corporate tax aggressiveness. Both the *BTDs* and *RBTDs* were scaled by the book value of total assets (Desai and Dharmapala, 2006; Chen et al., 2010).

3.2.2 Independent variable

The compensation contracts of both the CEO as well as the CFO were separately taken into account for the executive compensation variable (*EXEC_COMP*). The focus on the compensation of the CEO is justified given his influence on setting the corporate tone with respect to the tax activities of a firm (Gaertner, 2014). Also, studying the compensation contract of the CFO is relevant given the

fact that he has a greater responsibility with respect to the accounting function, as compared to the CEO (Rego and Wilson, 2008). According to Dyreng et al. (2010), both CEOs as well as CFOs significantly affect the tax avoidance activity of a firm.

Both the variable (V_COMP) and the fixed components (F_COMP) of the compensation of CEOs and the CFOs were separately considered for analysis. The V_COMP was measured as the sum of bonus, commission, and the Black-Scholes value of the stock options granted to the executives, as a percentage of the total compensation. F_COMP was taken as the natural logarithm of the sum of all other components of pay including salary, perquisites, contribution to provident fund and retirement benefits.

3.2.3 Moderating and control variables

Following previous studies, the percentage of institutional shareholdings was used to construct the moderating proxy for corporate governance strength (S_GOV) (Desai and Dharmapala, 2006; 2009; Drake et al., 2017; Herron and Nahata, 2018; Huang et al., 2018; Wahab and Holland, 2012). A dummy variable was formed that took a value of 1 in case the percentage of institutional shareholding was above the sample median and 0 otherwise (Huang et al., 2018). Since institutional shareholders play a crucial role in actively monitoring the management, a high percentage of such shareholdings indicates strong governance and vice versa (Jaiswall and Firth, 2016; Moore, 2012).

In order to capture the executive compensation - tax avoidance relationship adequately, several variables were taken into consideration to control for individual company characteristics following an extant literature review (e.g., Campbell et al., 2017; Chen et al., 2010; Gaertner, 2014; Halioui et al., 2016; Kiesewetter and Manthey, 2018; Phillips, 2003; Richardson et al., 2013). The variables controlled for include the size, profitability, leverage, capital intensity, growth, foreign activity, R&D intensity, operating volatility, and industry as well as year indicators.

Large size firms may attract more political attention and hence, this discourages them from being tax aggressive (Zimmerman, 1983). On the other hand, firms of large size may also lobby more frequently through which they may receive more favourable tax treatments and hence are

more tax aggressive (Porcano, 1986). Company size (*SIZE*) was determined as the natural logarithm of the book value of total assets (Phillips, 2003). Firms that have a huge profitability lie in high tax brackets and hence pay more taxes (Gupta and Newberry, 1997; Wilkie and Limberg, 1993). However, profitable firms also have a greater availability of resources and can exploit them for corporate tax aggressiveness (Rego, 2003). Profitability (*ROA*) was computed as the percentage of pre-tax income to total assets (Richardson et al., 2013).

Prior literature suggests that a high leverage is a sign of greater financial complexity (Mills et al., 1998). Also, these firms may be in a better position to be tax aggressive because of more such opportunities available to them (Newberry and Dhaliwal, 2001). On the other hand, there might be a substitution effect between the tax shield of debt and corporate tax shelters meaning that firms already having a large tax shield of debt are less likely to engage in corporate tax aggressiveness. Leverage (*LEV*) was computed as the ratio of the total book value of debt to assets (Moore, 2012). Capital intensive firms exhibit lower effective tax rates since they generate more tax credits (Gupta and Newberry, 1997). Capital intensity (*PPE*) was measured as the ratio of the net value of property, plant and equipment to total assets (Halioui et al., 2016).

Growth firms have a greater capability to take up large tax credits such as the allowance for depreciation since they make large capital expenditures and hence may have lower effective tax rates (Chen et al., 2010). In contrast, such firms are less experienced in corporate tax aggressiveness and so may exhibit higher effective tax rates (Campbell et al., 2017). The market to book ratio, computed as the ratio of market to book value of equity, was used to proxy for the firm growth (*MB*). Multinational corporations have a greater opportunity for engaging in tax aggressiveness since they have access to a number of jurisdictions and might shift income to a low tax country to be able to avoid taxes (Rego, 2003). Foreign activity (*FGN*) was calculated as the ratio of foreign income to total assets (Gaertner, 2014).

Firms that spend a lot of money on research and development can have lower taxes owing to the tax credits available to such firms on account of the R&D spending (Wilson, 2009). R&D

intensity (*RD*) was determined as the ratio of R&D expenditure to total assets (Kiesewetter and Manthey, 2018). Firms that are characterised by a greater variability in their earnings have an underlying motivation to engage in corporate tax aggressiveness (Dhaliwal et al., 2000; Phillips, 2003). Operating volatility (*STDEV*) was estimated as the standard deviation of the ROA over five years ending in the current year (Gaertner, 2014). Finally, Industry dummies (*IND*) and year dummies (*YR*) were also included in order to account for the industry and year fixed-effects (Armstrong et al., 2012; Campbell et al., 2017). A summary description of the variables has been provided in Table 3.

Table 3: Variable Description

Variable	Acronym	Description
Panel A: Corporate Tax Aggressiveness		
(1) GAAP Effective Tax rate	<i>GAAP_ETR</i>	Ratio of total tax to pre-tax income
(2) Current Effective Tax Rate	<i>CURR_ETR</i>	Ratio of current tax to pre-tax income
(3) Cash Effective Tax Rate	<i>CASH_ETR</i>	Ratio of taxes paid in cash to pre-tax income
(4) Book-Tax Difference	<i>BTD</i>	Difference, between book income and estimated taxable income, scaled by book value of total assets
(5) Residual Book-Tax Difference	<i>RBTD</i>	Residuals, from regression of total accruals on BTDs, scaled by book value of total assets
Panel B: Executive Compensation		
(1) Variable Compensation	<i>V_COMP</i>	Bonus, commission and equity compensation to CEO/CFO
(2) Fixed Compensation	<i>F_COMP</i>	All components of CEO/CFO pay excluding bonus, commission and equity compensation
Panel C: Corporate Governance Strength & Company-Specific Characteristics		
(1) Corporate Governance Strength	<i>S_GOV</i>	Binary variable; 1 if institutional shareholding is greater than sample median, 0 otherwise
(2) Size	<i>SIZE</i>	Natural log of book-value of total assets
(3) Profitability	<i>ROA</i>	Ratio of pre-tax income to total assets
(4) Leverage	<i>LEV</i>	Ratio of total debt to total assets
(5) Capital Intensity	<i>PPE</i>	Ratio of net property, plant and equipment to total assets
(6) Growth	<i>MB</i>	Ratio of market value to book value of equity
(7) Foreign Activity	<i>FGN</i>	Ratio of foreign income to total assets
(8) R&D Intensity	<i>RD</i>	Ratio of R&D expenditure to total assets
(9) Operating Volatility	<i>STDEV</i>	Standard deviation of ROA of 5 years ending in current year

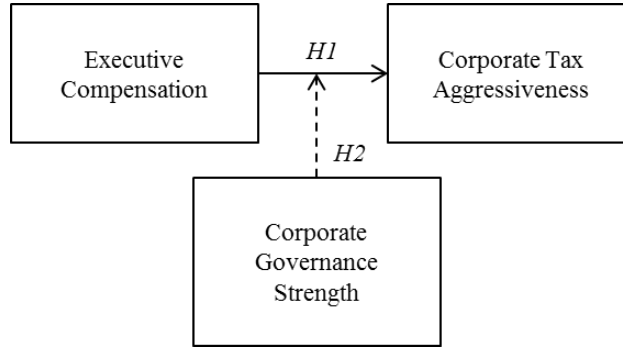
Source: Drawn from literature

3.3 Data Analysis

The conceptual framework relating to the present study has been shown in Figure 1. For the multivariate analysis, the impact of outliers was annulled by winsorising the variables at the 1 per cent level. Further, the variance inflation factors were checked in order to rule out the problem of

multicollinearity in the dataset (Mansfield and Helms, 1982). These came out to be below the rule-of-thumb value, i.e., 4 (O'brien, 2007). The software package STATA (Version 15) was used to carry out the data analysis.

Figure 1: Conceptual Framework



Source: Authors' depiction.

Multivariate panel data models were constructed to analyse the relationship between executive compensation and corporate tax aggressiveness. In order to decide whether the random- or fixed-effects model was appropriate, the Hausman test was carried out. The p -value of chi-square was found to be significant ($p < 0.05$) and hence, the fixed-effects model was used. Further, the Durbin-Watson test and the Breusch-Pagan test indicated the existence of autocorrelation and heteroskedasticity, respectively (Breusch and Pagan, 1979; Durbin and Watson, 1950). Hence, robust standard errors clustered at the firm level were employed. The fixed-effects panel regression equations are as follows:

$$TAXAGG = \alpha + \beta_1 EXEC_COMP + \beta_2 CONTROLS_{i,t} + \varepsilon \quad (1)$$

$$TAXAGG = \alpha + \beta_1 EXEC_COMP + \beta_2 S_GOV + \beta_3 EXEC_COMP * S_GOV + \beta_4 CONTROLS_{i,t} + \varepsilon, \quad (2)$$

$TAXAGG$ is alternatively $GAAP_ETR$, $CURR_ETR$, $CASH_ETR$, BTD or $RBTD$. $EXEC_COMP$ is either V_COMP or F_COMP of the CEO/CFO. $CONTROLS$ include $SIZE$, LEV , PPE , FGN , RD , ROA , MB , $STDEV$, IND , and YR .

4 Results and Discussion

Table 4 presents the summary statistics for the variables under study. As shown in the table, the mean (median) percentage of the variable component of CEO compensation was low at 34 (31) per

cent as compared to the fixed component with a mean (median) of 66 (69) per cent. Further, in case of the CFO, the mean (median) percentage of the variable component was even lower at 8 (0) per cent as compared to the fixed component with a mean (median) of 92 (100) per cent. Stock compensation, i.e., the value of stock options, as a percentage of the total compensation, was on average just 2 per cent in case of CEOs and a meagre 0.03 per cent in case of CFOs. The mean/median of all the effective tax rate measures (26 – 29 per cent) was found to be below the highest calculated statutory tax rates (33.99 – 34.94 per cent) over the study period, thus highlighting the presence of tax aggressiveness.

Table 4: Summary Statistics

Variables	(1)			(2)		
	Mean	Median	St. Dev.	Mean	Median	St. Dev.
Variable Compensation (Rs. million)	40.90	12.14	105.20	3.58	0.00	17.11
Fixed Compensation (Rs. million)	37.58	24.69	66.39	16.97	12.63	15.93
Total Compensation (Rs. million)	78.49	45.65	126.40	20.55	14.14	26.75
Stock Compensation (Rs. million)	6.62	0.00	74.59	1.27	0.00	12.71
Variable Compensation (% of total compensation)	34.00	31.00	32.00	0.08	0.00	0.17
Fixed Compensation (% of total compensation)	66.00	69.00	32.00	0.92	1.00	0.17
Stock Compensation (% of total compensation)	2.00	0.00	10.00	0.03	0.00	0.10
<i>V_COMP</i>	0.34	0.31	0.32	0.08	0.00	0.17
<i>F_COMP</i>	16.93	17.02	1.33	16.30	16.35	0.87
<i>GAAP_ETR</i>	-0.28	-0.29	0.09	-0.28	-0.29	0.09
<i>CURR_ETR</i>	-0.27	-0.27	0.09	-0.27	-0.27	0.09
<i>CASH_ETR</i>	-0.27	-0.26	0.10	-0.27	-0.26	0.10
<i>BTD</i>	0.03	0.02	0.04	0.03	0.02	0.04
<i>RBTD</i>	0.001	-0.004	0.03	0.001	-0.005	0.03
<i>S_GOV</i>	0.52	1.00	0.50	0.50	1.00	0.50
<i>SIZE</i>	10.49	10.36	1.21	10.55	10.42	1.23
<i>ROA</i>	0.13	0.11	0.09	0.13	0.11	0.09
<i>LEV</i>	0.33	0.31	0.18	0.33	0.30	0.18
<i>PPE</i>	0.25	0.24	0.15	0.25	0.24	0.16
<i>MB</i>	5.38	3.97	5.04	5.25	3.94	4.89
<i>FGN</i>	0.20	0.07	0.27	0.19	0.07	0.27
<i>RD</i>	0.01	0.001	0.02	0.01	0.001	0.02
<i>STDEV</i>	0.03	0.02	0.04	0.03	0.02	0.04

Source: Authors' compilation.

Notes: (1) Columns (1) and (2) denote the statistics calculated using the sample derived for CEO compensation and CFO compensation respectively as shown in Table 1. (2) Total compensation is the sum of variable compensation and fixed compensation. (3) Stock compensation is the Black-Scholes value of the stock options granted to the executives. (4) Other variables have been defined in Table 3.

Table 5 reports the correlation matrix for the variables under study. As shown in the table, there is a significant positive association between *V_COMP* and *BTD* as well as *RBTD* in

Table 5: Correlation Matrix

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
(1) <i>V_COMP</i>		0.10***	0.05*	0.06*	0.05	0.09***	0.08***	0.12***	0.12***	0.10***	-0.07**	-0.03	-0.03	0.07**	0.00	0.13***
(2) <i>F_COMP</i>	-0.20***		0.00	-0.01	-0.02	-0.01	0.00	0.27***	0.51***	0.01	-0.04	-0.04	0.10***	-0.15***	0.14***	-0.05*
(3) <i>GAAP_ETR</i>	-0.03	-0.05*		0.69***	0.49***	0.51***	0.50***	0.07**	0.12***	-0.06**	0.06*	-0.05	-0.06**	0.11***	0.13***	0.08**
(4) <i>CURR_ETR</i>	-0.01	-0.02	0.69***		0.65***	0.64***	0.64***	0.10***	0.15***	-0.07**	0.10***	0.11***	-0.11***	0.10***	0.11***	0.07**
(5) <i>CASH_ETR</i>	0.03	-0.02	0.47***	0.62***		0.44***	0.45***	0.07**	0.10***	-0.06*	0.14***	0.15***	-0.11***	0.07**	0.08***	0.05
(6) <i>BTD</i>	0.06*	0.01	0.50***	0.63***	0.43***		0.98***	0.06*	0.03	0.39***	-0.18***	0.00	0.10***	0.17***	0.22***	0.15***
(7) <i>RBTD</i>	0.05*	0.01	0.49***	0.63***	0.44***	0.98***		0.06**	0.03	0.33***	-0.14***	0.06**	0.08***	0.14***	0.21***	0.11***
(8) <i>S_GOV</i>	0.04	0.13***	0.09***	0.09***	0.06**	0.04	0.04		0.34***	-0.03	-0.04	-0.04	-0.04	0.01	-0.02	-0.08**
(9) <i>SIZE</i>	0.09***	0.31***	0.12***	0.15***	0.09***	-0.01	0.00	0.33***		-0.18***	0.15***	-0.04	-0.18***	0.03	0.03	-0.14***
(10) <i>ROA</i>	0.09***	0.02	-0.07**	-0.08***	-0.05*	0.39***	0.33***	-0.02	-0.19***		-0.48***	-0.16***	0.41***	0.30***	0.16***	0.35***
(11) <i>LEV</i>	-0.03	-0.03	0.07**	0.12***	0.13***	-0.18***	-0.14***	-0.04	0.16***	-0.47***		0.15***	-0.07**	-0.13***	-0.14***	-0.11***
(12) <i>PPE</i>	0.05	0.06**	-0.05*	0.10***	0.14***	-0.02	0.05	-0.09***	-0.06*	-0.16***	0.16***		-0.10***	-0.15***	-0.01	-0.02
(13) <i>MB</i>	-0.10***	0.01	-0.07**	-0.11***	-0.11***	0.10***	0.09***	-0.03	-0.17***	0.45***	-0.07**	-0.09***		-0.05	0.05*	0.08**
(14) <i>FGN</i>	0.07**	-0.11***	0.09***	0.09***	0.08***	0.13***	0.11***	0.00	0.00	0.26***	-0.12***	-0.14***	-0.06**		0.22***	0.14***
(15) <i>RD</i>	0.06**	0.08**	0.13***	0.11***	0.08***	0.24***	0.22***	-0.02	0.03	0.17***	-0.15***	-0.01	0.05	0.23***		0.20***
(16) <i>STDEV</i>	0.02	-0.02	0.07**	0.07**	0.04	0.16***	0.12***	-0.08**	-0.15***	0.35***	-0.12***	-0.02	0.08***	0.14***	0.20***	

Notes: (1) The lower and the upper triangular matrices show the correlations calculated using the sample obtained for CEO compensation and CFO compensation respectively as shown in Table 1.

(2) ***, ** and * indicate significance at the 1, 5 and 10 per cent levels respectively.

(3) Variables have been defined in Table 3.

Source: Authors' calculation.

case of CEOs, and between *V_COMP*, *GAAP_ETR*, *CURR_ETR*, *BTD* and *RBTD* in case of CFOs. A significant negative association can be seen between *F_COMP* and *GAAP_ETR* in case of CEOs. Also, *S_GOV* has a significant positive relationship with *GAAP_ETR*, *CURR_ETR* and *CASH_ETR* in case of CEOs, and with all the tax aggressiveness proxies in case of CFOs.

Table 6 portrays the results of the effect of variable and fixed compensation of the CEO on the alternative surrogates of corporate tax aggressiveness represented by the odd- and even-numbered models, respectively. No significant results were found for *V_COMP*. The results for *F_COMP*, however, highlight that there is a significant negative impact on *GAAP_ETR* (Model 2: $\beta = -0.005$; $p < 0.05$) and *CURR_ETR* (Model 4: $\beta = -0.004$; $p < 0.10$).

Table 7 presents the results of the moderating influence of corporate governance strength on the CEO compensation – corporate tax aggressiveness link. The results for the fixed component of pay, i.e., *S_GOV*F_COMP* show a significant positive influence on *CURR_ETR* (Model 4: $\beta = 0.012$; $p < 0.10$), *CASH_ETR* (Model 6: $\beta = 0.012$; $p < 0.10$), and *RBTD* (Model 10: $\beta = 0.003$; $p < 0.10$). No significant results were obtained for *S_GOV*V_COMP*. The results for *F_COMP* and *S_GOV* were significant for all the models, though negative for the former and positive for the latter. Amongst the controls, *MB* was found to be significantly negatively related in all models, except *GAAP_ETR*.

Table 8 displays the results of the impact of variable and fixed compensation in case of the CFO on the alternative proxies of tax aggressiveness represented by the odd- and even-numbered models, respectively. Similar to the results found in case of CEO compensation, *V_COMP* did not turn out to be significant in any of the models. Also, a significant negative impact of *F_COMP* was found in case of *BTD* (Model 8: $\beta = -0.002$; $p < 0.10$) as well as *RBTD* (Model 10: $\beta = -0.002$; $p < 0.10$).

Table 6: Impact of CEO Compensation on Corporate Tax Aggressiveness

Models	GAAP_ETR		CURR_ETR		CASH_ETR		BTD		RBTB	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>V_COMP</i>	-0.016 (0.012)		-0.008 (0.012)		-0.005 (0.012)		0.000 (0.005)		0.000 (0.005)	
<i>F_COMP</i>		-0.005** (0.002)		-0.004* (0.002)		-0.003 (0.002)		-0.001 (0.001)		-0.001 (0.001)
<i>SIZE</i>	0.008** (0.004)	0.010** (0.004)	0.008** (0.004)	0.010** (0.004)	0.004 (0.004)	0.005 (0.004)	0.001 (0.001)	0.001 (0.002)	0.001 (0.001)	0.002 (0.002)
<i>ROA</i>	-0.049 (0.056)	-0.056 (0.055)	-0.026 (0.057)	-0.026 (0.056)	0.104* (0.057)	0.105* (0.057)	0.236*** (0.041)	0.237*** (0.040)	0.215*** (0.041)	0.217*** (0.040)
<i>LEV</i>	0.021 (0.027)	0.015 (0.027)	0.043 (0.027)	0.040 (0.027)	0.084*** (0.031)	0.081*** (0.031)	0.017 (0.011)	0.017 (0.011)	0.021** (0.011)	0.021* (0.011)
<i>PPE</i>	-0.007 (0.033)	-0.005 (0.034)	0.059* (0.033)	0.061* (0.034)	0.082** (0.034)	0.084** (0.035)	0.020 (0.015)	0.021 (0.016)	0.032** (0.015)	0.033** (0.016)
<i>MB</i>	-0.001 (0.001)	-0.001 (0.001)	-0.002** (0.001)	-0.002** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.001** (0.001)	-0.001** (0.001)	-0.001* (0.001)	-0.001* (0.001)
<i>FGN</i>	-0.026 (0.022)	-0.029 (0.020)	-0.012 (0.025)	-0.015 (0.024)	-0.012 (0.017)	-0.015 (0.017)	-0.025* (0.015)	-0.026* (0.015)	-0.027* (0.016)	-0.028* (0.016)
<i>RD</i>	0.727* (0.390)	0.803** (0.369)	0.778** (0.370)	0.825** (0.360)	0.313 (0.363)	0.353 (0.357)	0.273* (0.157)	0.279* (0.157)	0.289* (0.156)	0.297* (0.156)
<i>STDEV</i>	0.177** (0.072)	0.184** (0.076)	0.178** (0.079)	0.181** (0.083)	0.063 (0.081)	0.065 (0.084)	-0.007 (0.040)	-0.007 (0.041)	-0.030 (0.045)	-0.029 (0.046)
Constant	-0.336*** (0.039)	-0.271*** (0.044)	-0.371*** (0.040)	-0.327*** (0.044)	-0.342*** (0.044)	-0.299*** (0.053)	-0.012 (0.017)	-0.001 (0.018)	-0.040** (0.017)	-0.026 (0.018)
Observations	1,063	1,063	1,063	1,063	1,063	1,063	1,063	1,063	1,063	1,063
R-squared	0.227	0.229	0.236	0.237	0.195	0.197	0.385	0.386	0.358	0.360
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: (1) Robust standard errors in parentheses; (2) ***, ** and * indicate significance at the 1, 5 and 10 per cent levels respectively; (3) Regression includes industry and year fixed-effects.

Source: Authors' analysis.

Table 7: Impact of Corporate Governance Strength on the relationship between CEO Compensation on Corporate Tax Aggressiveness

Models	GAAP_ETR		CURR_ETR		CASH_ETR		BTD		RBTB	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>V_COMP</i>	0.013 (0.018)		0.007 (0.018)		0.004 (0.017)		0.003 (0.007)		0.002 (0.007)	
<i>F_COMP</i>		-0.009** (0.005)		-0.008** (0.003)		-0.008*** (0.003)		-0.002* (0.001)		-0.002** (0.001)
<i>S_GOV</i>	0.017* (0.009)	0.018** (0.009)	0.023*** (0.008)	0.023*** (0.008)	0.020** (0.009)	0.020** (0.009)	0.007** (0.003)	0.007** (0.003)	0.007** (0.003)	0.007** (0.003)
<i>S_GOV*V_COMP</i>	-0.053 (0.023)		-0.032 (0.022)		-0.019 (0.024)		-0.007 (0.008)		-0.007 (0.008)	
<i>S_GOV*F_COMP</i>		0.006 (0.006)		0.012* (0.006)		0.012* (0.007)		0.003 (0.002)		0.003* (0.002)
<i>SIZE</i>	0.006 (0.004)	0.006 (0.004)	0.005 (0.004)	0.004 (0.005)	0.000 (0.005)	0.000 (0.005)	-0.000 (0.002)	-0.000 (0.002)	-0.000 (0.002)	-0.000 (0.002)
<i>ROA</i>	-0.054 (0.055)	-0.045 (0.053)	0.006 (0.055)	0.016 (0.054)	0.098* (0.054)	0.110** (0.054)	0.207*** (0.031)	0.210*** (0.029)	0.197*** (0.034)	0.199*** (0.033)
<i>LEV</i>	0.024 (0.026)	0.023 (0.027)	0.071** (0.031)	0.068** (0.031)	0.094*** (0.030)	0.090*** (0.029)	0.014 (0.009)	0.013 (0.009)	0.020** (0.010)	0.019** (0.009)
<i>PPE</i>	-0.016 (0.032)	-0.017 (0.034)	0.053 (0.034)	0.053 (0.036)	0.077** (0.033)	0.077** (0.034)	0.018 (0.013)	0.018 (0.013)	0.029** (0.013)	0.029** (0.014)
<i>MB</i>	-0.001 (0.001)	-0.001 (0.001)	-0.002** (0.001)	-0.002** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.001** (0.000)	-0.001** (0.000)	-0.001** (0.000)	-0.001** (0.000)
<i>FGN</i>	-0.015 (0.024)	-0.023 (0.022)	-0.023 (0.025)	-0.026 (0.024)	-0.014 (0.017)	-0.017 (0.016)	-0.013 (0.010)	-0.014 (0.009)	-0.015 (0.010)	-0.016 (0.010)
<i>RD</i>	0.664* (0.386)	0.783** (0.364)	0.657 (0.398)	0.739* (0.387)	0.292 (0.359)	0.369 (0.353)	0.304** (0.150)	0.319** (0.149)	0.314** (0.148)	0.333** (0.146)
<i>STDEV</i>	0.149** (0.074)	0.176** (0.073)	0.224 (0.189)	0.226 (0.193)	0.097 (0.180)	0.085 (0.182)	-0.009 (0.035)	-0.007 (0.035)	-0.031 (0.041)	-0.029 (0.042)
Constant	-0.331*** (0.041)	-0.183** (0.082)	-0.357*** (0.046)	-0.223*** (0.078)	-0.321*** (0.047)	-0.184** (0.078)	-0.005 (0.017)	0.025 (0.025)	-0.033* (0.017)	0.004 (0.025)
Observations	1,063	1,063	1,063	1,063	1,063	1,063	1,063	1,063	1,063	1,063
R-squared	0.226	0.221	0.227	0.229	0.205	0.209	0.432	0.434	0.402	0.404
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: (1) Robust standard errors in parentheses; (2) ***, ** and * indicate significance at the 1, 5 and 10 per cent levels respectively; (3) Regression includes industry and year fixed-effects.

Source: Authors' analysis.

Table 9 shows the results of the moderating influence of corporate governance strength on the relationship between CFO compensation and corporate tax aggressiveness. These were also very similar to the results found in case of CEO compensation. A significant positive impact of $S_GOV * F_COMP$ was found on all the proxies of tax aggressiveness except $GAAP_ETR$. No significant results were obtained for $S_GOV * V_COMP$. With the exception of one of the models, the results for F_COMP and S_GOV were significant, though negative for the former and positive for the latter. Amongst the controls, MB was again found to be significantly negatively related in all models, except $GAAP_ETR$.

Overall, the results do not provide evidence of a significant association between the variable component of executive compensation and tax aggressiveness. Thus, $H1a$ stands rejected. This might be due to a relatively insignificant contribution of the variable pay to the total executive compensation as earlier shown in Table 4. Further, the performance-based stock options are meagrely, rendering the executive pay insensitive to stock return volatility. The executives, therefore, don't have the necessary motivation to engage in risky tax-aggressive activities which would have otherwise increased the volatility of company's stock returns and therefore, the value of the stock option portfolios held by them. Thus, in India, the variable executive compensation does not seem to be contributing to the incentive effect as put forth by other researchers (Balasubramanian et al., 2010; Chakrabarty et al., 2012; Jaiswall et al., 2016). The findings, however, document a significant negative relationship between the fixed component of executive compensation and corporate tax aggressiveness, thus upholding $H1b$. These findings are in line with the analysis done by Ghosh (2006) and Jaiswall and Firth (2009) suggesting a negative relationship between executive compensation and risk-taking in the case of corporate India. Further, the mostly cash-based fixed component of an executive compensation in India can be implicated as the underlying basis

Table 8: Impact of CFO Compensation on Corporate Tax Aggressiveness

Models	GAAP_ETR		CURR_ETR		CASH_ETR		BTD		RBTD	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>V_COMP</i>	-0.005 (0.020)		0.003 (0.022)		0.001 (0.024)		0.000 (0.008)		0.001 (0.007)	
<i>F_COMP</i>		-0.002 (0.006)		-0.005 (0.005)		-0.003 (0.006)		-0.002* (0.001)		-0.002* (0.001)
<i>SIZE</i>	0.008** (0.003)	0.008* (0.004)	0.009*** (0.003)	0.011** (0.004)	0.005 (0.004)	0.006 (0.005)	0.002*** (0.001)	0.003*** (0.001)	0.002** (0.001)	0.003*** (0.001)
<i>ROA</i>	-0.086 (0.053)	-0.087* (0.052)	-0.043 (0.057)	-0.042 (0.055)	0.099 (0.061)	0.099* (0.059)	0.229*** (0.029)	0.229*** (0.029)	0.210*** (0.028)	0.210*** (0.028)
<i>LEV</i>	0.007 (0.027)	0.007 (0.026)	0.030 (0.027)	0.030 (0.027)	0.084*** (0.030)	0.083*** (0.030)	0.013 (0.009)	0.013 (0.009)	0.017** (0.008)	0.017** (0.008)
<i>PPE</i>	-0.002 (0.033)	-0.003 (0.033)	0.074** (0.032)	0.074** (0.032)	0.077** (0.034)	0.077** (0.034)	0.030*** (0.009)	0.029*** (0.009)	0.041*** (0.009)	0.041*** (0.009)
<i>MB</i>	-0.001 (0.001)	-0.001 (0.001)	-0.001* (0.001)	-0.001* (0.001)	-0.003*** (0.001)	-0.003** (0.001)	-0.001** (0.000)	-0.001* (0.000)	-0.001* (0.000)	-0.001* (0.000)
<i>FGN</i>	-0.019 (0.023)	-0.020 (0.022)	-0.007 (0.027)	-0.012 (0.025)	-0.015 (0.020)	-0.018 (0.020)	-0.022** (0.009)	-0.024** (0.010)	-0.024** (0.010)	-0.026*** (0.010)
<i>RD</i>	0.776** (0.394)	0.804** (0.397)	0.777** (0.374)	0.846** (0.371)	0.315 (0.407)	0.353 (0.410)	0.268*** (0.098)	0.294*** (0.098)	0.278*** (0.101)	0.304*** (0.101)
<i>STDEV</i>	0.207*** (0.071)	0.205*** (0.071)	0.184** (0.080)	0.190** (0.080)	0.073 (0.081)	0.075 (0.082)	0.002 (0.029)	0.004 (0.029)	-0.022 (0.032)	-0.019 (0.031)
Constant	-0.337*** (0.037)	-0.314*** (0.076)	-0.374*** (0.038)	-0.309*** (0.071)	-0.351*** (0.042)	-0.314*** (0.075)	-0.027** (0.011)	0.001 (0.019)	-0.054*** (0.011)	-0.026 (0.019)
Observations	1,029	1,029	1,029	1,029	1,029	1,029	1,029	1,029	1,029	1,029
R-squared	0.242	0.242	0.243	0.244	0.211	0.212	0.383	0.385	0.360	0.362
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: (1) Robust standard errors in parentheses; (2) ***, ** and * indicate significance at the 1, 5 and 10 per cent levels respectively; (3) Regression includes industry and year fixed-effects.

Source: Authors' analysis.

Table 9: Impact of Corporate Governance Strength on the relationship between CFO Compensation and Corporate Tax Aggressiveness

Models	GAAP_ETR		CURR_ETR		CASH_ETR		BTD		RBTD	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>V_COMP</i>	-0.005 (0.021)		-0.004 (0.023)		-0.004 (0.025)		-0.003 (0.006)		-0.003 (0.006)	
<i>F_COMP</i>		-0.003 (0.007)		-0.015** (0.007)		-0.012* (0.007)		-0.003** (0.002)		-0.004** (0.002)
<i>S_GOV</i>	0.013 (0.009)	0.013 (0.009)	0.024*** (0.009)	0.025*** (0.009)	0.018* (0.009)	0.019** (0.009)	0.007*** (0.002)	0.007*** (0.002)	0.007*** (0.002)	0.007*** (0.002)
<i>S_GOV*V_COMP</i>	0.001 (0.006)		0.005 (0.006)		0.009 (0.007)		0.001 (0.001)		0.001 (0.002)	
<i>S_GOV*F_COMP</i>		0.003 (0.008)		0.016** (0.008)		0.018** (0.009)		0.004* (0.002)		0.004** (0.002)
<i>SIZE</i>	0.006 (0.004)	0.006 (0.005)	0.004 (0.004)	0.007 (0.005)	0.000 (0.005)	0.003 (0.005)	0.001 (0.001)	0.002 (0.001)	0.001 (0.001)	0.001 (0.001)
<i>ROA</i>	-0.079 (0.054)	-0.080 (0.053)	-0.000 (0.058)	0.000 (0.055)	0.097 (0.061)	0.096* (0.058)	0.205*** (0.020)	0.205*** (0.020)	0.192*** (0.021)	0.192*** (0.020)
<i>LEV</i>	0.011 (0.027)	0.011 (0.027)	0.059** (0.030)	0.060** (0.030)	0.092*** (0.029)	0.093*** (0.029)	0.010 (0.006)	0.010* (0.006)	0.016** (0.006)	0.016** (0.006)
<i>PPE</i>	-0.012 (0.033)	-0.014 (0.033)	0.067** (0.033)	0.062* (0.034)	0.071** (0.034)	0.067** (0.034)	0.027*** (0.008)	0.025*** (0.008)	0.038*** (0.008)	0.036*** (0.008)
<i>MB</i>	-0.001 (0.001)	-0.000 (0.001)	-0.002* (0.001)	-0.001 (0.001)	-0.003*** (0.001)	-0.003** (0.001)	-0.001*** (0.000)	-0.001** (0.000)	-0.001** (0.000)	-0.001** (0.000)
<i>FGN</i>	-0.008 (0.025)	-0.009 (0.025)	-0.020 (0.027)	-0.029 (0.026)	-0.016 (0.020)	-0.023 (0.020)	-0.011* (0.006)	-0.013** (0.006)	-0.012** (0.006)	-0.014** (0.006)
<i>RD</i>	0.724* (0.390)	0.755* (0.387)	0.627 (0.401)	0.772** (0.385)	0.237 (0.405)	0.360 (0.388)	0.296*** (0.092)	0.328*** (0.091)	0.297*** (0.093)	0.331*** (0.093)
<i>STDEV</i>	0.204*** (0.072)	0.202*** (0.072)	0.286 (0.191)	0.293 (0.190)	0.165 (0.180)	0.171 (0.178)	0.005 (0.026)	0.005 (0.025)	-0.017 (0.029)	-0.017 (0.028)
Constant	-0.323*** (0.043)	-0.284*** (0.100)	-0.348*** (0.047)	-0.145 (0.104)	-0.314*** (0.048)	-0.143 (0.103)	-0.015 (0.011)	0.033 (0.025)	-0.042*** (0.011)	0.009 (0.025)
Observations	1,029	1,029	1,029	1,029	1,029	1,029	1,029	1,029	1,029	1,029
R-squared	0.230	0.230	0.229	0.235	0.221	0.226	0.437	0.439	0.409	0.412
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: (1) Robust standard errors in parentheses; (2) ***, ** and * indicate significance at the 1, 5 and 10 per cent levels respectively; (3) Regression includes industry and year fixed-effects.

Source: Authors' analysis.

for this risk-averse behavior, as was also concluded by Rego and Wilson (2012) in case of US firms and by Huang et al. (2018) in case of Chinese firms.

Finally, no significant impact of the strength of corporate governance on the relationship between the variable component of executive compensation and tax aggressiveness was found, thus rejecting *H2a*. However, the study did find a significant positive moderating effect of governance strength on the negative relationship between fixed part of executive compensation and corporate tax aggressiveness, thus lending support to *H2b*. Given the fact that tax aggressiveness adds value to the firm in the long run, such positive relationship of a robust governance environment stands justified (Minnick and Noga, 2010).

In order to check for robustness of the results, alternate specifications were tested. First, the analysis was conducted taking the total compensation of the executives as the independent variable. Results found were similar to the results reported in case of fixed compensation of the executives. Second, using the lagged value of total assets as a deflator also did not change the previously reported results. Finally, the analysis was conducted by including firm fixed effects instead of the industry fixed effects and again, similar results were obtained.

5 Conclusion

The emerging literature on corporate tax avoidance examines the same within the framework of an agency setting. Corporate tax aggressiveness is analogous to a risky project that has a positive net present value and hence is favoured by the shareholders but not by the risk fearing managers (Rego and Wilson, 2012). This necessitates the need to incentivise the managers to take up the required risk in order to enhance shareholders' wealth (Crocker and Slemrod, 2005). The vast expanse of literature on the incentive effect of managerial compensation contracts on corporate tax aggressiveness in the developed economies,

primarily the US, has majorly focused on equity-based compensation. These studies have suggested that equity-based incentives do motivate the managers to engage in risky tax aggressive positions owing to the relationship between such risky tax strategies and the volatility of stock returns which, in turn, increases the value of their stock option portfolios (Armstrong et al., 2012, 2015; Minnick and Noga, 2010; Rego and Wilson, 2012). These findings, however, cannot be generalised to emerging economies like India where the variable component of compensation, especially equity-based pay, is not very prominent (Chakrabarty et al., 2012; Balasubramanian et al., 2010). In order to bridge this research gap, the present study explored the relationship between executive compensation and corporate tax aggressiveness in India by taking into account the compensation of both the CEOs and the CFOs separately for a subset of firms in the S&P BSE 500 Index during the period from 1 April, 2014 to 31 March, 2019. Further, the moderating effect of the strength of corporate governance on the relationship between executive compensation and corporate tax aggressiveness, as proposed by Desai and Dharmapala (2006), was also tested.

The descriptive analysis of the sample companies highlighted a low average variable to fixed compensation proportion of 34:66 in case of Indian CEOs and an even lower proportion of 8:92 in case of Indian CFOs. The value of stock options granted to Indian executives was just 2 per cent of the total pay in case of CEOs and a measly 0.03 per cent in case of CFOs. Also, both mean as well median effective tax rates were found below the statutory tax rates, thus, revealing the prevalence of corporate tax aggressiveness in India. The econometric analysis showcased a significant negative relationship between the fixed component of executive compensation and tax aggressiveness. Further, a significant moderating effect of the strength of corporate governance was established on the relationship between the fixed part of executive compensation and corporate tax aggressiveness. However, none of the hypothesised conjectures for the link between the variable component

of executive compensation and tax aggressiveness, with and without the moderation of corporate governance strength, were validated.

The results of the study indicate the prevalence of the risk-related agency conundrum in case of India. Executive compensation in India is skewed in favour of the fixed component as compared to the variable component. Thus, contrary to the executive pay in case of developed economies where the variable pay constitutes most of the compensation paid, there is a clear lack of adequate incentives to the Indian executives to engage in risky tax aggressive positions. Further, since the bulk of executive compensation is cash-based, executives tend to have a greater level of risk aversion as documented by Rego and Wilson (2012). This is consistent with the findings of previous Indian studies highlighting a negative relationship between executive compensation and risk-taking (Ghosh, 2006; Jaiswall and Firth, 2009). Huang et al. (2018) also reported similar results in case of China.

The findings of this study provide valuable insights to the practitioners and policy-makers highlighting the relevance of variable pay in aligning the incentives of the managers and the shareholders. This work extends the existing literature on corporate tax aggressiveness that has a western skew to it. The results should, however, be interpreted with care as they would be relevant in the context of institutional settings comparable to that of India and are not generalisable to the rest of the world. Future research efforts can focus on the compensation contracts of other executives and other emerging economies. The moderating impact of alternative governance structures on the executive compensation – corporate tax aggressiveness link can be evaluated, furthering research in this domain.

6 References

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